

What do concepts mean in foresight, really?

Twenty working definitions

Scenario

- A description of a possible or plausible path from the present to some future state.
- Constructed by holding a specified set of present conditions fixed, varying others, and tracing the causal dependencies that follow.
- Three things belong to the definition itself. They are not add-ons.

(1) A scenario is always a member of a set

- A single scenario is a prediction in disguise, whatever it is called.
- The set exists in order to display the structure of the possibility space: what depends on what, and where the paths fork.
- For this reason "which one do you believe?" is not a question a scenario set can answer. It is the wrong question to ask.

One scenario out of the deck, believed, is a forecast wearing a different hat.

(2) Its success condition is not that it comes true

- Most scenarios in a set will not occur; that is the design, not a failure.
- A scenario succeeds if it identifies correctly the dependencies, the mechanisms behind them, and the genuine difference-makers in the system.
- It can be literally false about the endpoint and still deliver understanding, in the same way as a frictionless plane does in physics.

Shell's 1973 scenario was not believed in the boardroom. It did its job anyway.

(3) Its engine is counterfactual reasoning

- What if X rather than Y? Every scenario is an answer to a question of this form, whether the question is written down or not.
- A good scenario isolates the factors that would make a difference from those that would not.
- This is what separates a scenario from a story, however well the story is told.

Take the war out and ask what changes. If nothing does, it was a story.

Possible

- A future that nothing we know rules out.
- One needs to state which kind of possibility is meant: logical (not self-contradictory), physical (allowed by the laws of nature), or practical (consistent with the world as it now stands, its institutions and resources included).
- These are different things, and one should not switch between them when challenged.

A Mars colony is possible. So is your lottery win. Say which kind you mean.

Plausible

- A possible future for which a causal pathway from the present can be sketched, step by step, with the mechanisms named.
- The absence of identified obstacles is not yet plausibility; something has to be said about how we get there, not only that nothing stops us.

Nobody has proven you cannot walk to the moon. That is not a travel plan.

Probable

- Ill-formed until the grounds are stated.
- The grounds may be a model, structured expert judgment with documented reasoning, or individual credence. These are not equally authoritative, and the reader is owed to know which is in use.
- Where no grounds can be given responsibly, the term does not apply, and "likely" is no escape.

"70 percent" from a model and "70 percent" from a man at the bar are the same number and a different claim.

Desirable

- A future judged good or just under some value framework, and at least possible.
- If there is no pathway, we have an ideal without a strategy.
- If there is no agent whose choices could bring it about, we have a wish rather than a target.

A carbon-neutral city with nobody to build it is a postcard.

Forecast

- A conditional claim about what will probably happen, concerning delimited phenomena whose structure is known well enough to be modelled.
- The condition matters: a forecast holds given that the modelled structure holds.
- Evaluated by its correspondence with the outcome – unlike a scenario.

The weatherman can be wrong. That is exactly why we listen to him.

Vision

- A description of a future worth aiming at.
- Normative by definition: it says what ought to be pursued, not what will be.
- Reading it as a description of what will be is a category error, and a common one.

A vision of a walkable city is not a claim about 2040. It is an instruction for Monday.

Weak signal

- An early indication that some development is possible, before anything can be said about its plausibility or probability.
- Interpretation-laden from the start: what counts as a signal depends on what one is already prepared to see.
- There is no such thing as a raw signal.

Three teenagers doing something odd is noise. Three teenagers and a theory is a signal.

Time horizon

- How far ahead the question is extended.
- The horizon decides which futures enter the discussion in the first place, and which are ruled out before anyone has argued against them.
- The choice is therefore substantive, not merely practical.

Ask about 2030 and fusion is off the table. Ask about 2060 and it sets the table.

Method

- A procedure together with its presuppositions: where knowledge of the future is located, who counts as knowing, and what kind of future the method can yield.
- These presuppositions are rarely stated in the manual, but they do their work all the same.
- When we choose a method, we inherit its presuppositions, whether we notice it or not.

A Delphi panel assumes the experts know. A citizens' jury assumes they do not. Pick one and you have picked a theory of knowledge.

Understanding

- The capacity to move in the space of possibilities: to grasp the dependency structure well enough to answer what-if questions that have not yet been encountered.
- Non-factive: one can understand a system well and still be wrong about what it does next.
- This, and not prediction, is the primary epistemic aim of the field, or so I argue.

You do not know where the ball lands. You do know what the wind does to it.

Prediction

- A claim that a specified state will obtain at a specified time, evaluated by correspondence with the outcome.
- Legitimate for delimited and well-modelled phenomena; a category mistake as the general aim of the field.
- Much of the criticism of foresight is criticism of prediction aimed at the wrong target.

Tide tables predict. Foresight asks what you do with the tide.

Counterfactual

- A claim about what would have happened, or would happen, had things been otherwise.
- The unit of causal reasoning in historiography and in foresight alike: to say that X mattered is to say that without X, things would have gone differently.
- Every scenario is built out of counterfactuals, whether or not the word is used.

No Sarajevo, no war? The question is not idle. It is the whole claim.

Difference-maker

- A factor whose variation would change the outcome. Every cause is one; that is what being a cause amounts to.
- The question is never whether something makes a difference, but to what, against which contrast, and at which level of description. Oxygen makes a difference to every fire; it makes no difference to why this fire started rather than not.
- A good scenario picks out the difference-makers that answer the question asked, and leaves the rest in the background where they belong.

Oxygen made a difference to every fire in history. Nobody calls the fire brigade about the oxygen.

Possibility space

- The set of futures not ruled out under a stated kind of possibility, together with their dependency structure.
- It is this, and not any single member of it, that a scenario set is about.
- The space is not given; it is constructed by the choices of horizon, method, and what is held fixed.

A chess engine does not tell you the opponent's next move. It tells you the board.

Conceivability

- What can be thought or imagined as a future, here and now.
- Historically variable and socially constrained: what one age cannot conceive, the next takes for granted.
- The limits of conceivability are not the limits of possibility, and confusing the two is a recurring error of the field.

In 1900 nobody could conceive the internet. The internet did not mind.

Presentism

- The view that only the present exists: the past is gone and the future is nothing yet.
- It grounds the open future the field assumes, but it makes it harder to say what futures knowledge is knowledge of, if there is no future to be known.
- A position the field has largely taken without arguing for it.

There is no 2040 to visit. There are only the 2020s, which is where 2040 is being made.

Futures knowledge

- Knowledge of present structures, dependencies, and possibilities that bear on the future – not knowledge of the future itself.
- The future is not observed; it is inferred from what is already the case.
- This is why the quality of futures knowledge is the quality of one's grasp of the present.

Nobody has seen 2040. Plenty of people have seen the birth statistics.

Trend

- A regularity in past data extended forward.
- Only as good as the stability of the mechanism behind it, which the trend itself does not display.
- To extend a trend is to make a claim about a mechanism; that claim needs its own support.

The chicken's growth curve looked excellent every morning, including the last one.

Wild card

- An event of low estimated probability and high impact.
- The category is epistemic, not ontological: it marks where the dependency structure is poorly grasped, not a special type of event in the world.
- Many wild cards were foreseeable; they were simply outside the horizon or the method.

The pandemic was on every risk register. The surprise was that nobody had read it.

Strict enough to disagree with

- These definitions are strict, and one may well disagree with many of them. That is, in a sense, the point.
- A definition that cannot be disagreed with does not do any work.
- What remains open is whether the field is willing to have the disagreement at all.

Define the word first, then fight about it. The other order wastes everybody's afternoon.